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the solute) is directly proportional to the mole fraction(x) of the gaseous solute in the solution at low concentrations". This statement is known as Henry's law.

Henry's law can be expressed as,

$$p_{solute} \alpha x_{solute\ in\ solution} \quad (9.1)$$

$$p_{solute} = K_H x_{solute\ in\ solution} \quad (9.2)$$

Here, p_{solute} represents the partial pressure of the gas in vapour state which is commonly called as vapour pressure. x_{solute} in solution represents the mole fraction of solute in the solution. K_H is a empirical constant with the dimensions of pressure. The value of 'K_H' depends on the nature of the gaseous solute and solvent. The above equation is a straight-line in the form of y=mx. The plot partial pressure of the gas against its mole fraction in a solution will give a straight line as shown in fig 9.3. The slope of the straight line gives the value of K_H.

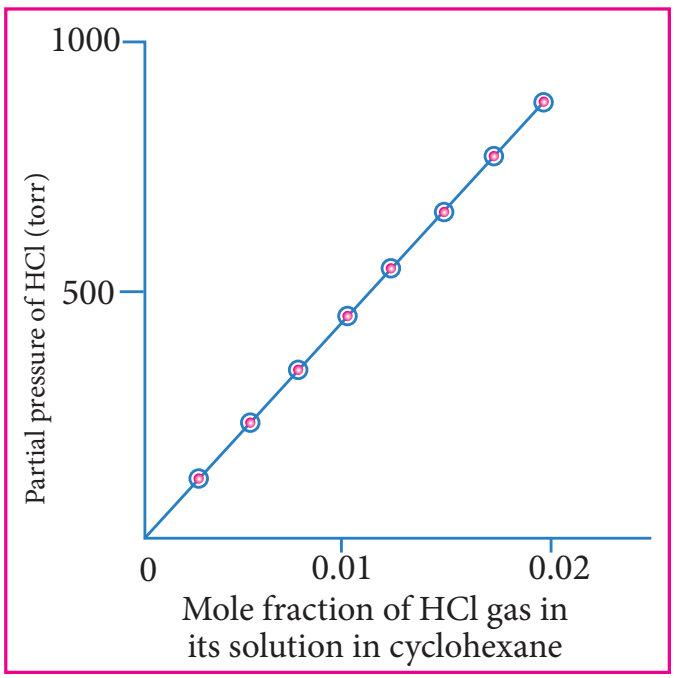

Figure 9.3 Solubility of HCl gas in cyclohexane at 293 K.

Why the carbonated drinks are stored in a pressurized container?

We all know that the carbonated beverages contain carbon dioxide dissolved in them. To dissolve the carbon dioxide in these drinks, the CO_2 gas is bubbled through them under high pressure. These containers are sealed to maintain the pressure. When we open these containers at atmospheric pressure, the pressure of the CO_2 drops to the atmospheric level and hence bubbles of CO_2 rapidly escape from the solution and show effervescence. The burst of bubbles is even more noticeable, if the soda bottle is in warm condition.